# Discrete Mathematics — Unit 2: Mathematical Logic

A proposition is a statement that is either true or false. Connectives: ¬, ∧, ∨, → and ↔. p → q is false only when p is true and q is false; it is equivalent to ¬p ∨ q.

## Tautologies and equivalences

A tautology is true under every assignment, e.g. p ∨ ¬p. Useful equivalences: contrapositive (p → q ≡ ¬q → ¬p), De Morgan, distribution, absorption.

## Predicates and quantifiers

∀x P(x) and ∃x P(x). Negation: ¬∀x P(x) ≡ ∃x ¬P(x). Rules of inference: modus ponens, modus tollens, hypothetical syllogism, universal instantiation.